

JavaScript Operators



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01

Introduction

What are Operators?

Operators are special symbols or keywords used to perform operations on **variables** and **values**.

JavaScript uses **operators** to perform:

Calculations

Comparisons

Logical decisions

Assignments of values





02

Arithmetic Operators

Arithmetic Operators

Operator	Description	Example	Result
+	Addition	$5 + 2$	7
-	Subtraction	$5 - 2$	3
*	Multiplication	$5 * 2$	10
/	Division	$5 / 2$	2.5
%	Modulus (Remainder)	$5 \% 2$	1
**	Exponentiation	$2 ** 3$	8
++	Increment	<code>a++</code>	<code>a = a + 1</code>
--	Decrement	<code>a--</code>	<code>a = a - 1</code>





03

Comparison Operators

Comparison Operators

Operator	Description	Example	Result
==	Equal to	5 == '5'	true
===	Strict equal (type + value)	5 === '5'	false
!=	Not equal	5 != '5'	false
!==	Strict not equal	5 !== '5'	true
>	Greater than	5 > 2	true
<	Less than	5 < 2	false
>=	Greater than or equal	5 >= 5	true
<=	Less than or equal	5 <= 4	false





04

Assignment Operators

Assignment Operators

Operator	Example	Equivalent To
=	<code>a = 5</code>	assign 5 to a
<code>+=</code>	<code>a += 2</code>	<code>a = a + 2</code>
<code>-=</code>	<code>a -= 2</code>	<code>a = a - 2</code>
<code>*=</code>	<code>a *= 2</code>	<code>a = a * 2</code>
<code>/=</code>	<code>a /= 2</code>	<code>a = a / 2</code>
<code>%=</code>	<code>a %= 2</code>	<code>a = a % 2</code>





05

Logical Operators

Logical Operators

Operator	Description	Example	Result
&&	Logical AND	true && false	FALSE
!	Logical NOT	!true	FALSE
	Logical OR	true false	TRUE



Difference between Null and Undefined



Feature	null	undefined
Type	Object (typeof null === "object")	Undefined (typeof undefined === "undefined")
Meaning	Intentional absence of value	Variable declared but not assigned
Set by	Developer (manually assigned)	JavaScript (automatically)
Use Case	To clear or reset a value	To check if a value is uninitialized
Example	let a = null;	let b; or let b = undefined;

